MERCK MANUAL
Professional Version

Chronic Obstructive Pulmonary Disease (COPD)

(Chronic Obstructive Bronchitis; Emphysema)

*Full Review: Jul 2026 By **Robert A. Wise**, MD, Johns Hopkins Asthma and Allergy Center | Peer reviewed by **M. Patricia Rivera**, MD, University of Rochester Medical Center*
Last updated: Jul 2026

Chronic obstructive pulmonary disease (COPD) is airflow limitation caused by an inflammatory response to inhaled toxins, often cigarette smoke. Alpha-1 antitrypsin deficiency and various occupational exposures are less common causes in patients who do not smoke. Symptoms are productive cough and dyspnea that develop over years; common signs include decreased breath sounds, prolonged expiratory phase of respiration, and wheezing. Severe cases may be complicated by weight loss, pneumothorax, frequent acute decompensation episodes, right heart failure, and/or acute or chronic respiratory failure. Diagnosis is based on history, physical examination, chest radiograph, and pulmonary function tests. Treatment is with bronchodilators, glucocorticoids, and, when necessary, oxygen and antibiotics. Biologics may be used for selected patients with eosinophilic inflammation and frequent exacerbations. Lung volume reduction procedures or transplantation are used in advanced disease. Survival in COPD is related to the severity of airflow limitation, the frequency of exacerbations, and the presence of comorbidities.

Chronic obstructive pulmonary disease (COPD) is defined as a heterogeneous lung condition characterized by chronic respiratory symptoms (dyspnea, cough, sputum production, and/or exacerbations) due to abnormalities of the airways (bronchitis, bronchiolitis) and/or alveoli (emphysema) that cause persistent, often progressive, airflow obstruction ([1]).

COPD comprises:

- Chronic obstructive bronchitis (clinically defined)
- Emphysema (pathologically or radiologically defined)

Many patients have features of both.

Chronic obstructive bronchitis is chronic bronchitis with airflow obstruction. Chronic bronchitis is defined as productive cough on most days of the week for a total of at least 3 months per year for 2 consecutive years. Chronic bronchitis becomes chronic obstructive bronchitis if spirometric evidence of airflow obstruction develops. Chronic asthmatic bronchitis is a similar, overlapping condition characterized by chronic productive cough, wheezing, and partially reversible airflow obstruction; it occurs predominantly in patients who smoke and have a history of asthma. When asthma and COPD co-exist in the same patient, pharmacologic treatment should follow guidelines for asthma (1).

Emphysema is destruction of lung parenchyma leading to a loss of elastic recoil, loss of alveolar septa, and radial airway traction, which increases the tendency for airway collapse. Lung hyperinflation, airflow limitation, and air trapping follow. Airspaces enlarge and may eventually develop blebs or bullae. Obliteration of small airways is thought to be the earliest lesion that precedes the development of emphysema.

Asthma and COPD can coexist in the
same patient, a clinical presentation sometimes described as asthma-COPD overlap.

General reference

1. Global Initiative for Chronic Obstructive Lung Disease (GOLD): Diagnosis and assessment. Global Strategy for the Prevention, Diagnosis, and Management of COPD: 2026 report.

Epidemiology of COPD

The prevalence, incidence, and mortality rates of COPD increase with age. COPD may exhibit familial clustering independent of alpha-1 antitrypsin deficiency (alpha-1 antiprotease inhibitor deficiency).

Estimates of the global prevalence of COPD vary widely depending on the method of ascertainment. For example, the Global Burden of Disease study reported a prevalence of COPD of 2.5% in 2021 (1). In the United States, the National Health and Nutritional Examination Survey (NHANES) found a prevalence of airflow limitation in 13.4% of adults, although 71% of these individuals had not been diagnosed with COPD (2). In the National Health Interview Survey, the prevalence of self-reported COPD was higher in women (4.1%) than men (3.4%) (3).

COPD remains a leading cause of death worldwide (1, 4, 5, 6). Approximately 141,000 deaths from COPD occur annually in the United States (7).

Epidemiology references

1. Wang Z, Lin J, Liang L, et al. Global, regional, and national burden of chronic obstructive pulmonary disease and its attributable risk factors from 1990 to 2021: an analysis for the Global Burden of Disease Study 2021. *Respir Res.*2025;26(1):2. doi:10.1186/s12931-024-03051-2
2. Martinez CH, Mannino DM, Jaimes FA, et al. Undiagnosed Obstructive Lung Disease in the United States. Associated Factors and Long-term Mortality. *Ann Am Thorac Soc.* 2015;12(12):1788-1795. doi:10.1513/AnnalsATS.201506-388OC
3. Weeks JD, Elgaddal N. Chronic Obstructive Pulmonary Disease in Adults Age 18 and Older: United States, 2023. *NCHS Data Brief.* 2025;(529):1. doi:10.15620/cdc/174596

4. GBD 2023 Causes of Death Collaborators. Global burden of 292 causes of death in 204 countries and territories and 660 subnational locations, 1990-2023: a systematic analysis for the Global Burden of Disease Study 2023. *Lancet.* 2025;406(10513):1811-1872. doi:10.1016/S0140-6736(25)01917-8
5. World Health Organization (WHO). Fact Sheet. Chronic obstructive pulmonary disease (COPD). November 6, 2024. Accessed April 30, 2026.
6. Centers for Disease Control and Prevention. National Center for Health Statistics: Leading Causes of Death. Updated February 5, 2026. Accessed April 29, 2026.
7. Centers for Disease Control and Prevention. Chronic Obstructive Pulmonary Disease (COPD). Updated May 15, 2024. Accessed April 29, 2026.

Etiology of COPD

There are 2 main causes of COPD:

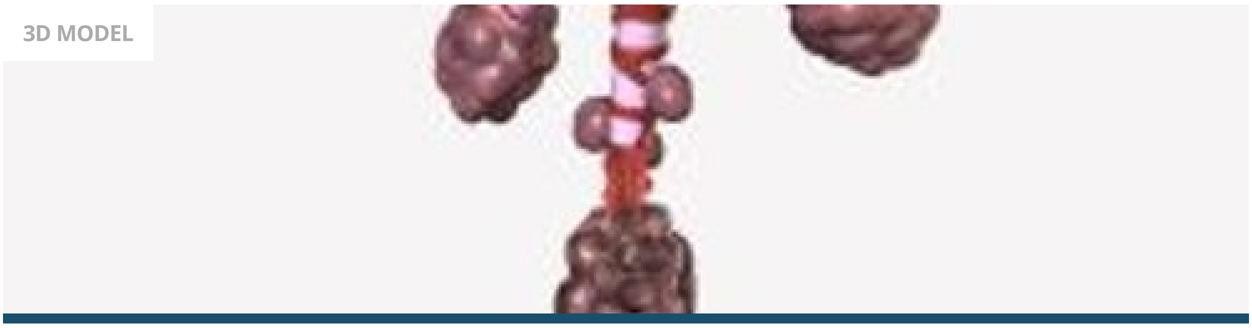
- Smoking (and less often other inhalational exposures)
- Genetic factors

Smoking and other inhalational exposures

Among all inhalational exposures, cigarette smoking is the primary risk factor in most countries, although only approximately 15% of people who smoke develop clinically apparent COPD (1). The risk for COPD increases with both duration (years of smoking) and cumulative dose (pack-years) (2). People who smoke and also have preexisting airway reactivity (defined by increased sensitivity to inhaled methacholine), even in the absence of clinically apparent asthma, are at greater risk of developing COPD than are those without airway reactivity.

Alveoli and Bronchiole Damage Resulting From Smoking

3D MODEL



Smoke from indoor cooking and heating is an important causative factor in countries where indoor fires are commonly used for cooking or heating (3). The magnitude of risk of development of COPD from biomass cooking is highest in resource-limited countries and may be confounded by the other effects of poverty (4).

Exposure to passive cigarette smoke, air pollution, and occupational dust (eg, mineral dust, cotton dust) or inhaled chemicals (eg, cadmium) also contribute to the risk of COPD but are of less relevance compared to cigarette smoking. Low body weight and childhood respiratory disorders (eg, childhood wheezing, asthma) also contribute to the risk of COPD.

Genetic factors

The best-defined causative genetic disorder is alpha-1 antitrypsin deficiency, which is an important cause of emphysema in people who do not smoke and markedly increases susceptibility to development of COPD in people who do.

Over 80 genetic alleles have been found to be associated with COPD or decline in lung function in selected populations (5). However, the most consequential allele is alpha-1 antitrypsin.

Etiology references

1. Wheaton AG, Liu Y, Croft JB, et al. Chronic Obstructive Pulmonary Disease and Smoking Status - United States, 2017. *MMWR Morb Mortal Wkly Rep.* 2019;68(24):533-538. doi:10.15585/mmwr.mm6824a1
2. Bhatt SP, Kim YI, Harrington KF, et al. Smoking duration alone provides stronger risk estimates of chronic obstructive pulmonary disease than pack-years. *Thorax.* 2018;73(5):414-421. doi:10.1136/thoraxjnl-2017-210722
3. Ortiz-Quintero B, Martínez-Espinosa I, Pérez-Padilla R. Mechanisms of Lung Damage and Development of COPD Due to Household Biomass-Smoke Exposure: Inflammation, Oxidative Stress, MicroRNAs, and Gene Polymorphisms. *Cells.* 2022;12(1):67. doi:10.3390/cells12010067
4. Mortimer K, Montes de Oca M, Salvi S, et al. Household air pollution and COPD: cause and effect or confounding by other aspects of poverty? *Int J Tuberc Lung Dis.* 2022;26(3):206-216. doi:10.5588/ijtld.21.0570
5. Cho MH, Hobbs BD, Silverman EK. Genetics of chronic obstructive pulmonary disease: understanding the pathobiology and heterogeneity of a complex disorder. *Lancet Respir Med.* 2022;10(5):485-496. doi:10.1016/S2213-2600(21)00510-5

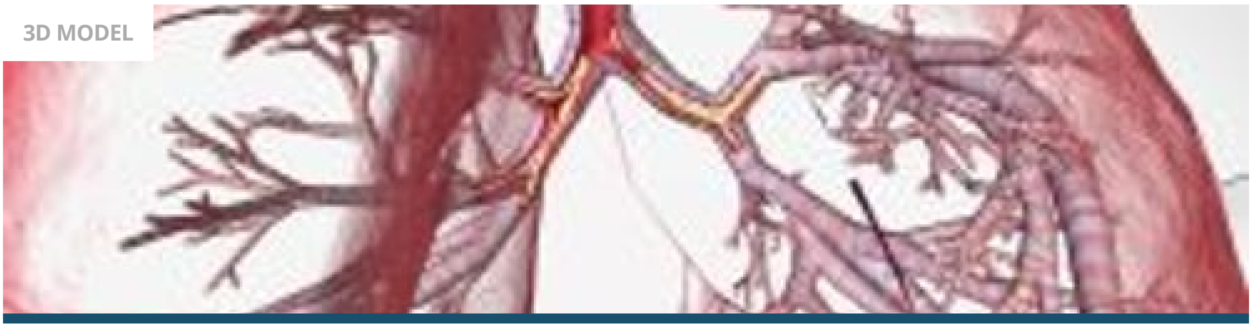
Pathophysiology of COPD

Various factors contribute to the pathophysiology of COPD, including a complex interplay of chronic inflammation, oxidative damage, and infection, leading to structural remodeling and progressive, incompletely reversible airflow limitation.

Inflammation

Inhalational exposures can trigger an inflammatory response in airways and alveoli that leads to disease in genetically susceptible people. The process is thought to be mediated by an increase in protease activity and a decrease in antiprotease activity. Lung proteases, such as neutrophil elastase, matrix metalloproteinases, and cathepsins, break down elastin and connective tissue during normal tissue repair. Their activity is normally balanced by antiproteases, such as alpha-1 antitrypsin, airway epithelium–derived secretory leukoproteinase inhibitor, elafin, and matrix metalloproteinase tissue inhibitor. In patients with COPD, activated neutrophils and other inflammatory cells (eg, macrophages) release proteases as part of the inflammatory process; protease activity exceeds antiprotease activity, and tissue destruction and mucus hypersecretion result.

Bronchitis



Activation of neutrophils and macrophages also leads to oxidative damage via the accumulation of free radicals, superoxide anions, and hydrogen peroxide, which inhibit antiproteases and cause bronchoconstriction, mucosal edema, and mucous hypersecretion. Neutrophil-induced oxidative damage, release of profibrotic neuropeptides (eg, bombesin), and reduced levels of vascular endothelial growth factor (VEGF) may contribute to apoptotic destruction of lung parenchyma.

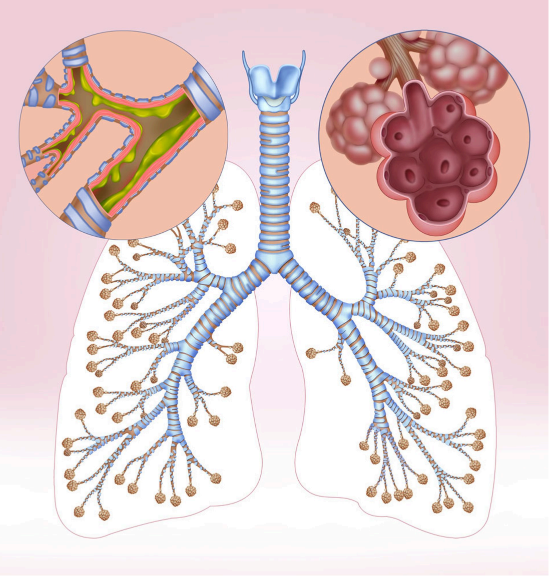
The inflammation in COPD increases as disease severity increases, and, in severe (advanced) disease, inflammation does not resolve completely despite smoking cessation. This chronic inflammation may not respond to glucocorticoids, particularly in patients who continue to smoke cigarettes (1).

Emphysema

Mechanical stress on the alveoli from over-distension may make them susceptible to proteases, leading to alveolar septal destruction and progressive emphysema. This is particularly notable in the upper lobes of the lung (2).

Emphysema and Bronchitis (Illustration)

IMAGE



This illustration of the lungs and bronchial branches shows bronchitis and emphysema. In emphysema (inset top right), the alveoli are destroyed, affecting lung elasticity. Chronic bronchitis (inset top left) is characterized by the persistent inflammation and mucus in the bronchi.

ILUSMEDICAL/SCIENCE PHOTO LIBRARY

Infection

Respiratory infections (to which patients with COPD are prone) may amplify progression of lung destruction.

Bacteria, especially Haemophilus influenzae, colonize the lower airways of approximately 30% of patients with COPD (3). In more severely affected patients (eg, those with previous hospitalizations), colonization with Pseudomonas aeruginosa or other gram-negative bacteria is common. Smoking and airflow obstruction may lead to impaired mucus clearance in lower airways, which predisposes to infection. Repeated bouts of infection increase the inflammatory burden that hastens disease progression. There is no direct evidence, however, that long-term use of antibiotics slows disease progression in COPD, despite the short-term benefits of some antibiotics (eg, azithromycin) in reducing exacerbation frequency and improving quality of life (4).

Viral infections, particularly rhinovirus, can account for up to 50% of COPD exacerbations and lead to progression of disease via intense airway epithelial inflammation (5).

Airflow limitation

The cardinal pathophysiologic feature of COPD is airflow limitation caused by airway narrowing and/or obstruction, loss of elastic recoil, or both.

There are 2 basic pathways by which COPD can develop and manifest with symptoms in later life:

- In the first pathway, patients may have normal lung function in early adulthood, which is followed by a more rapid decline in the volume of air forcefully expired during the first second after taking a full breath (FEV_1; ≥ 60 mL/year).
- In the second pathway, patients have impaired lung function in early adulthood, often associated with asthma or other childhood respiratory disease. In these patients, COPD may present with a normal age-related decline in FEV_1 (approximately 30 mL/year).

Although this second pathway model is conceptually helpful, a wide range of individual trajectories is possible (6). When the FEV_1 falls below approximately 1 L, patients develop dyspnea during activities of daily living (although dyspnea is more closely related to the degree of dynamic hyperinflation [progressive hyperinflation due to incomplete exhalation] than to the degree of airflow limitation). When the FEV_1 falls below approximately 0.8 L, patients are at risk of hypoxemia, hypercapnia, and cor pulmonale.

Airway narrowing and obstruction are caused by inflammatory mucus hypersecretion, mucus plugging, mucosal edema, bronchospasm, peribronchial fibrosis, and remodelling of small airways or a combination of these mechanisms. Alveolar septa are destroyed, reducing parenchymal attachments to the airways and thereby facilitating airway closure during expiration.

Enlarged alveolar spaces sometimes consolidate into bullae, defined as airspaces ≥ 1 cm in diameter. Bullae may be entirely empty or have strands of lung tissue traversing them in areas of locally severe emphysema; they occasionally occupy an entire hemithorax. These changes lead to loss of elastic recoil and lung hyperinflation.

Increased airway resistance increases the work of breathing. Lung hyperinflation, although it decreases airway resistance, also increases the work of breathing. Increased work of breathing may lead to respiratory muscle fatigue and alveolar hypoventilation with hypercapnia. Hypoxia and hypercapnia can also be caused by ventilation/perfusion (V/Q) mismatch.

Complications

In addition to airflow limitation and sometimes respiratory insufficiency, complications include:

- Pulmonary hypertension
- Respiratory infection
- Weight loss and other comorbidities

Chronic alveolar hypoxia increases pulmonary vascular tone, which, if diffuse, causes pulmonary hypertension and cor pulmonale. The increase in pulmonary vascular pressure may be augmented by the destruction of the pulmonary capillary bed due to destruction of alveolar septa.

Viral or bacterial respiratory infections are common among patients with COPD and cause a large percentage of acute exacerbations. It is currently thought that acute bacterial infections are due to acquisition of new strains of bacteria rather than overgrowth of chronic colonizing bacteria.

Weight loss may occur, and has been consistently associated with the insufficient caloric intake and increased levels of circulating tumor necrosis factor (TNF)-alpha that is seen in patients with COPD (7, 8).

A mismatch between caloric expenditure and nutritional intake typically occurs because caloric expenditure can be high in the presence of heightened inflammatory cytokines and hypoxemia.

Other coexisting or complicating disorders that adversely affect quality of life and/or survival include frailty, osteoporosis, depression, anxiety, coronary artery disease, arrhythmias, lung cancer and other cancers, muscle atrophy, and gastroesophageal reflux. Whether these disorders arise from COPD itself, smoking, or shared pathogenic mechanisms causing systemic inflammation remains uncertain.

Pathophysiology references

1. Adcock IM, Bhatt SP, Balkissoon R, Wise RA. The Use of Inhaled Corticosteroids for Patients with COPD Who Continue to Smoke Cigarettes: An Evaluation of Current Practice. *Am J Med.* 2022;135(3):302-312. doi:10.1016/j.amjmed.2021.09.006
2. Suki B, Sato S, Parameswaran H, Szabari MV, Takahashi A, Bartolák-Suki E. Emphysema and mechanical stress-induced lung remodeling. *Physiology (Bethesda).* 2013;28(6):404-413. doi:10.1152/physiol.00041.2013
3. Short B, Carson S, Devlin AC, et al: Non-typeable *Haemophilus influenzae* chronic colonization in chronic obstructive pulmonary disease (COPD). *Crit Rev Microbiol.* 2021;47(2):192-205. https://doi.org/10.1080/1040841X.2020.1863330
4. Janjua S, Mathioudakis AG, Fortescue R, et al. Prophylactic antibiotics for adults with chronic obstructive pulmonary disease: a network meta-analysis. *Cochrane Database Syst Rev.* 2021;1(1):CD013198. doi:10.1002/14651858.CD013198.pub2
5. Owuor N, Nalamala N, Gimenes JA Jr, Sajjan US. Rhinovirus and COPD airway epithelium. *Pulm Crit Care Med.* 2017;2(3):10.15761/PCCM.1000139. doi:10.15761/PCCM.1000139
6. Lange P, Celli B, Agusti A, et al: Lung-function trajectories leading to chronic obstructive pulmonary disease. *N Engl J Med* 373(2):111–122, 2015.
7. De Brandt J, Beijers RJHCG, Chiles J, et al. Update on the Etiology, Assessment, and Management of COPD Cachexia: Considerations for the Clinician. *Int J Chron Obstruct Pulmon Dis.* 2022;17:2957-2976. Published 2022 Nov 18. doi:10.2147/COPD.S334228
8. Takabatake N, Nakamura H, Abe S, et al. The relationship between chronic hypoxemia and activation of the tumor necrosis factor-alpha system in patients with chronic obstructive pulmonary disease. *Am J Respir Crit Care Med.* 2000;161(4 Pt 1):1179-1184. doi:10.1164/ajrccm.161.4.9903022

Symptoms and Signs of COPD

COPD takes years to develop and progress.

- Productive cough is the initial symptom, typically developing among people who smoke and are in their 40s and 50s.
- Dyspnea that is progressive, persistent, exertional, or worse during respiratory infection appears when patients are in their late 50s or 60s.

Other symptoms include chest tightness, fatigue, and activity limitation, all of which are more pronounced during an exacerbation. Symptoms usually progress quickly in patients who continue to smoke and in those who have a higher lifetime tobacco exposure. Morning headache develops in more advanced disease and signals nocturnal hypercapnia or hypoxemia.

Signs of COPD include wheezing, a prolonged expiratory phase of breathing, lung hyperinflation manifested as decreased heart and lung sounds, and increased anteroposterior diameter of the thorax (barrel chest).

Breathing with COPD

3D MODEL

Signs of advanced disease include pursed-lip breathing, accessory respiratory muscle use, paradoxical inward movement of the lower rib cage during inspiration (Hoover sign), and cyanosis. Signs of cor pulmonale include neck vein (eg, jugular vein) distention, splitting of the second heart sound with an accentuated pulmonic component, tricuspid insufficiency murmur, and peripheral edema. Right ventricular heaves may be imperceptible in COPD because the lungs are hyperinflated.

Patients with advanced emphysema lose weight and experience muscle wasting.

Spontaneous pneumothorax may occur (possibly related to rupture of bullae) and should be suspected in any patient with COPD whose pulmonary status abruptly worsens.

The symptoms can be graded according to which activities cause dyspnea (see table Breathlessness Measurement using the mMRC Questionnaire).

TABLE

Breathlessness Measurement Using the Modified British Medical Research Council (mMRC) Questionnaire

Grade	Shortness of Breath
0	None except during strenuous exercise
1	Occurring when hurrying on level ground or walking up a slight incline

Grade	Shortness of Breath
2	Resulting in walking more slowly than people of the same age on level ground *or* Resulting in stopping for breath when walking at own pace on level ground
3	Resulting in stopping for breath after walking about 100 meters or after a few minutes on level ground
4	Preventing the person from leaving the house *or* Occurring when dressing or undressing

Adapted from Mahler DA, Wells CK. Evaluation of clinical methods for rating dyspnea. *Chest.* 1988;93(3):580-586. doi: 10.1378/chest.93.3.580

Acute exacerbations

Acute exacerbations occur sporadically during the course of COPD and are heralded by increased symptom severity. The specific cause of any exacerbation is almost always impossible to determine, but exacerbations are often attributed to viral respiratory infections, acute bacterial bronchitis, or exposure to respiratory irritants. As COPD progresses, acute exacerbations tend to become more frequent, averaging approximately 1 to 3 episodes/year.

Diagnosis of COPD

- Chest imaging
- Pulmonary function testing

The diagnosis of COPD is suggested by a combination of history, physical examination, and chest imaging findings and is confirmed by pulmonary function tests (1).

Symptoms similar to those of COPD can be caused by asthma, heart failure, and bronchiectasis (see table Differential Diagnosis of COPD). COPD and asthma are sometimes easily confused and may coexist in an individual (also called asthma-COPD overlap).

TABLE

Differential Diagnosis of COPD

Diagnosis	Onset	Imaging Results	Other Features
COPD	Middle age	Sometimes lung hyperinflation, bullae, increased retrosternal air space, and/or bronchial wall thickening (seen on chest imaging); however, imaging is usually not helpful diagnostically and is done mainly to exclude other disorders	Slowly progressive symptoms History of smoking or exposure to tobacco or other types of smoke
Asthma	Early in life (often during childhood)	Usually normal or possibly hyperinflation (eg, during an asthma exacerbation)	Symptoms vary widely from day to day Variable but reversible airflow obstruction Symptoms often worse at night or early morning History of allergies, rhinosinusitis, or eczema Often family history of asthma
Bronchiectasis	All ages, but most often in older or middle aged adults	Bronchial dilation and bronchial wall thickening (seen on chest radiograph or chest CT)	Often large amounts of purulent sputum Often history of chronic recurrent bacterial infection

Diagnosis	Onset	Imaging Results	Other Features
Diffuse panbronchiolitis	Usually between ages 10 and 60 years (mean age of 40 years)	Diffuse small centrilobular nodular opacities and hyperinflation seen on chest radiograph and high-resolution CT	Mostly males who do not smoke Almost all patients have chronic sinusitis Predominately in individuals of Asian descent
Heart failure	All ages, but most often in older or middle aged adults	Enlarged heart, pleural effusion, fluid in major lung fissure, sometimes pulmonary edema (seen on chest radiograph)	Volume restriction without airflow limitation (detected by pulmonary function tests)
Obliterative bronchiolitis	Onset at younger age	Peripheral hypodense areas (seen on chest CT during expiration)	History of lung or bone marrow transplantation Inspiratory squeaks (ie, high-pitched, end-inspiratory wheezes) on auscultation
Tuberculosis (TB)	All ages	Lung infiltrates and cavitation (typically on the upper lobes) seen on chest radiograph	Confirmed by microbiologic testing (eg, NAAT, AFB staining, mycobacterial culture) Usually in areas with high prevalence of TB

Data adapted from Global Initiative for Chronic Obstructive Lung Disease (GOLD): Diagnosis and assessment. Global Strategy for the Prevention, Diagnosis, and Management of COPD: 2026 report.

AFB = acid-fast bacillus; NAAT = nucleic acid amplification test.

Systemic disorders that may have a component of airflow limitation may suggest COPD; they include HIV infection, abuse of illicit drugs (particularly cocaine and amphetamines), sarcoidosis, Sjögren syndrome, bronchiolitis obliterans, lymphangioleiomyomatosis, and eosinophilic granuloma. COPD can be differentiated from interstitial lung diseases by chest imaging, which shows increased interstitial markings in interstitial lung disease, and pulmonary function testing, which shows a restrictive rather than an obstructive ventilatory defect. In some patients, COPD and interstitial lung disease coexist (combined pulmonary fibrosis and emphysema [CPFE]) in which lung volumes are relatively preserved, but gas exchange is severely impaired.

Pulmonary function tests

Patients suspected of having COPD should undergo pulmonary function testing to confirm airflow limitation, to quantify its severity and reversibility, and to distinguish COPD from other disorders. Pulmonary function testing is also useful for following disease progression and monitoring response to treatment. The primary diagnostic parameters are:

- FEV_1: The volume of air forcefully expired during the first second after taking a full breath
- Forced vital capacity (FVC): The total volume of air expired with maximal force
- Flow-volume loops: Simultaneous spirometric recordings of airflow and volume during forced maximal expiration and inspiration

In patients with COPD, there are reductions of FEV_1, FVC, and the ratio of FEV_1/FVC are the hallmark of airflow limitation. Flow-volume loops show a concave pattern in the expiratory tracing (see figure Flow-Volume Loop).

FEV_1 and FVC are easily measured with office spirometry. Normal reference values are determined by patient age, sex, and height. It is recommended that race not be used for calculating predicted reference values (2). (See also Airflow, Lung Volumes, and Flow-Volume Loop.)

CLINICAL CALCULATORS

PFT predicted values by 2022 race-neutral GLI equations, ages 3 to 95

Airflow limitation severity in patients with COPD and FEV_1/FVC < 0.70 can be classified based on post-bronchodilator FEV_1 (1):

- Mild: ≥ 80% of predicted

- Moderate: 50% to 79% of predicted
- Severe: 30% to 49% of predicted
- Very severe: < 30% of predicted

Additional pulmonary function testing is necessary only in specific circumstances, such as before lung volume reduction procedures. Other test abnormalities may include:

- Increased total lung capacity
- Increased functional residual capacity
- Increased residual volume
- Decreased vital capacity
- Decreased single-breath diffusing capacity for carbon monoxide (DLCO)

Findings of increased total lung capacity, functional residual capacity, and residual volume can help distinguish COPD from restrictive pulmonary disease, in which these measures are diminished.

Decreased DLCO is nonspecific and is reduced in other disorders that affect the pulmonary vascular bed, such as interstitial lung disease, but can help distinguish emphysema from asthma, in which DLCO is normal or elevated.

A determination of whether a patient's symptoms are a result of COPD or asthma is best made by comparison of pulmonary function test findings (see table PFT Findings in COPD and Asthma).

TABLE

Pulmonary Function Test (PFT) Findings in Chronic Obstructive Pulmonary Disease (COPD) and Asthma

PFT Parameter	Typical COPD Finding	Typical Asthma Finding
FEV_1/FVC Ratio	< 0.70	Typically normal < 0.80 during asthma exacerbation
FEV_1	Little to no improvement with bronchodilator (< 12% and < 200 mL)	Significant improvement with bronchodilator (> 12% and > 200 mL)
RV	Increased (> 120–140%)	Normal Increased only during asthma exacerbation
TLC	Increased (> 120%)	Increased only during asthma exacerbation
DLCO	Decreased	Normal or increased

DLCO = diffusing capacity for carbon monoxide; FEV_1 = volume of air forcefully expired during the first second after taking a full breath; FVC = forced vital capacity; RV = residual volume ; TLC = total lung capacity.

Data adapted from Global Initiative for Chronic Obstructive Lung Disease (GOLD): Diagnosis and assessment. Global Strategy for the Prevention, Diagnosis, and Management of COPD: 2026 report and Global Initiative for Asthma (GINA). Global Strategy for Asthma Management and Prevention: 2026 report.

Imaging tests

Chest radiograph may reveal characteristic findings. In patients with emphysema, changes can include lung hyperinflation manifested as a flat diaphragm (ie, increase in the angle formed by the sternum and anterior diaphragm on a lateral film from the normal value of 45° to > 90°), rapid tapering of hilar vessels, and bullae (ie, radiolucencies > 1 cm surrounded by arcuate, hairline shadows). Other typical findings include enlargement of the retrosternal airspace and a narrow cardiac shadow. Emphysematous changes occurring predominantly in the lung bases suggest alpha-1 antitrypsin deficiency. The lungs may look normal or have increased lucency secondary to loss of parenchyma. Among patients with chronic obstructive bronchitis, chest radiographs may be normal or may show a bibasilar increase in bronchovascular markings as a result of bronchial wall thickening.

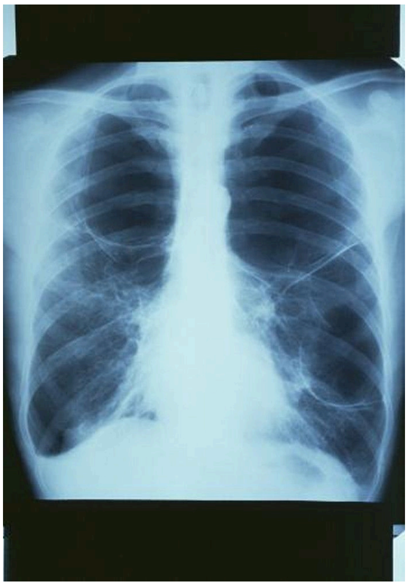
Chronic Obstructive Pulmonary Disease (Chest Radiograph)

IMAGE

Chest radiograph of a patient with COPD. The lungs are hyperinflated, the diaphragm is flattened, vascular markings are increased, and the heart size is marginally increased.

COPD with Bullae

IMAGE



This chest radiograph shows a large bulla in the upper right lung and 2 large bullae in the left lung.

GJLP/CNRI/SCIENCE PHOTO LIBRARY

Prominent hila suggest large central pulmonary arteries that may signify pulmonary hypertension. Right ventricular enlargement that occurs in cor pulmonale may be masked by lung hyperinflation or may manifest as encroachment of the heart shadow on the retrosternal space or by widening of the transverse cardiac shadow in comparison with previous chest radiographs.

Chest CT may reveal abnormalities that are not apparent on the chest radiograph and may also suggest coexisting or complicating disorders, such as pneumonia, pneumoconiosis, or lung cancer. CT helps assess the extent and distribution of emphysema, estimated either by visual scoring or with analysis of the distribution of lung density. Indications for obtaining CT in patients with COPD include evaluation for lung volume reduction procedures, suspicion of coexisting or complicating disorders that are not clearly evident or excluded by chest radiograph, suspicion of lung cancer, and screening for lung cancer. Enlargement of the pulmonary artery diameter greater than the ascending aorta diameter suggests pulmonary hypertension (3).

Adjunctive tests

Alpha-1 antitrypsin levels should be measured in patients < 50 years with symptomatic COPD and in patients of any age with COPD who do not smoke to detect alpha-1 antitrypsin deficiency. Other indications of possible alpha-1 antitrypsin deficiency include a family history of premature COPD or unexplained liver disease, lower-lobe distribution of emphysema, and COPD associated with antineutrophil cytoplasmic antibody (ANCA)-positive vasculitis. If levels of alpha-1 antitrypsin are low, the diagnosis should be confirmed by genetic testing to establish the alpha-1 antitrypsin phenotype.

ECG, often done to exclude cardiac causes of dyspnea, typically shows diffusely low QRS voltage with a vertical heart axis caused by lung hyperinflation and increased P-wave voltage or rightward shifts of the P-wave vector caused by right atrial enlargement in patients with advanced emphysema. Findings of right ventricular hypertrophy include an R or R' wave as tall as or taller than the S wave in lead V1; an R wave smaller than the S wave in lead V6; right-axis deviation > 110° without right bundle branch block; or some combination of these findings. Multifocal atrial tachycardia, an arrhythmia that can accompany COPD, manifests as a tachyarrhythmia with polymorphic P waves and variable PR intervals.

Echocardiography is occasionally useful for assessing right ventricular function and pulmonary hypertension, although air trapping makes it technically difficult in patients with COPD. Echocardiography is most often indicated when coexistent left ventricular or valvular heart disease is suspected.

Hemoglobin and hematocrit are of little diagnostic value in the evaluation of COPD but may show reactive erythrocytosis (hematocrit > 48%) if the patient has chronic hypoxemia. Patients with anemia (for reasons other than COPD) have disproportionately severe dyspnea. The differential white blood cell count (eg, neutrophilia, eosinophilia) may be helpful. Neutrophilia has been associated with a higher incidence of exacerbations and eosinophilia is well-established to predict response to inhaled glucocorticoids and anti-interleukin (IL)-5 biologic therapies.

Serum electrolytes are of little value but may show an elevated bicarbonate level (ie, compensatory metabolic alkalosis) if patients have chronic hypercapnia. Venous blood gases are useful for diagnosis of acute or chronic hypercapnia.

Evaluation of exacerbations

Patients with acute exacerbations usually have combination of increased cough, sputum, dyspnea, and work of breathing, as well as low oxygen saturation determined with pulse oximetry, diaphoresis, tachycardia, anxiety, and cyanosis. Patients with exacerbations accompanied by retention of carbon dioxide may be lethargic or somnolent.

All patients requiring hospitalization for an acute exacerbation should undergo testing to quantify hypoxemia and hypercapnia. Hypercapnia may exist without hypoxemia.

Findings of partial pressure of arterial oxygen (PaO_2) < 50 mm Hg, or partial pressure of carbon dioxide in arterial blood ($PaCO_2$) > 50 mm Hg, or partial pressure of carbon dioxide in venous blood ($PvCO_2$) > 55 mm Hg in patients with respiratory acidemia (pH < 7.35) define acute respiratory failure. Some patients chronically manifest such levels of PaO_2 and $PaCO_2$ in the absence of acute respiratory failure.

A chest radiograph is often done to check for pneumonia or pneumothorax. Point-of-care ultrasound (POCUS) has been increasingly useful as an adjunctive procedure for rapid bedside diagnosis of pneumonia or pneumothorax. In patients with acute onset of symptoms, chest CT angiogram is done to check for pulmonary embolism. Very rarely, among patients receiving chronic systemic glucocorticoids, infiltrates may represent Aspergillus pneumonia.

Yellow or green sputum may be associated with neutrophils (mainly, because of the presence of the enzyme myeloperoxidase) in the sputum and raises suspicion for the presence of bacterial colonization or infection (4).

Sputum culture is usually done in hospitalized patients but is not usually necessary in outpatients. In samples from outpatients, Gram stain usually shows neutrophils with a mixture of organisms, often gram-positive diplococci (*Streptococcus pneumoniae*), gram-negative bacilli (*H. influenzae*), or both. However, culture and microscopic examination of sputum is usually not necessary for outpatients. Other oropharyngeal commensal organisms, such as *Moraxella catarrhalis* (formerly known as *Branhamella catarrhalis*), occasionally cause exacerbations. In hospitalized patients, resistant gram-negative organisms (eg, *Pseudomonas*) or, rarely, *Staphylococcus* may be identified in culture specimens.

During influenza season, a rapid influenza test may help guide treatment with anti-influenza agents (eg, oseltamivir), and a respiratory viral panel for respiratory syncytial virus (RSV), rhinovirus, and metapneumovirus may allow tailoring of antimicrobial therapy. Testing for COVID-19 and consideration of COVID-19–specific therapies is also indicated.

Serum C-reactive protein (CRP) is helpful in guiding the use of antibiotics during exacerbations. In clinical trials, the use of antibiotics can be decreased in patients with low CRP without evidence of harm (5, 6). Evidence regarding use of the inflammatory biomarker procalcitonin is conflicting; while it has been found to be beneficial in guiding antibiotic therapy in some outpatient settings, meta-analyses have largely been inconclusive and ICU studies have shown harm (7). Therefore, it is not recommended to use procalcitonin to guide antibiotic treatment during COPD exacerbations.

Diagnosis references

1. Global Initiative for Chronic Obstructive Lung Disease (GOLD): Diagnosis and assessment. Global Strategy for the Prevention, Diagnosis, and Management of COPD: 2026 report.
2. Bhakta NR, Bime C, Kaminsky DA, et al. Race and ethnicity in pulmonary function test interpretation: An Official American Thoracic Society Statement. *Am J Respir Crit Care Med.* 2023;207(8):978-995. doi:10.1164/rccm.202302-0310ST
3. Iyer AS, Wells JM, Vishin S, et al. CT scan-measured pulmonary artery to aorta ratio and echocardiography for detecting pulmonary hypertension in severe COPD. *Chest.* 2014;145(4):824-832. doi:10.1378/chest.13-1422
4. Spies R, Potter M, Hollamby R, et al. Sputum Color as a Marker for Bacteria in Acute Exacerbations of Chronic Obstructive Pulmonary Disease: A Systematic Review and Meta-analysis. *Ann Am Thorac Soc.* 2023;20(5):738-748. doi:10.1513/AnnalsATS.202204-319OC
5. Butler CC, Gillespie D, White P, et al. C-Reactive protein testing to guide antibiotic prescribing for COPD exacerbations. *N Engl J Med.* 2019;381(2):111-120. doi: 10.1056/NEJMoa1803185
6. Prins HJ, Duijkers R, van der Valk P, et al. CRP-guided antibiotic treatment in acute exacerbations of COPD in hospital admissions. *Eur Respir J.* 2019;53(5):1802014. doi: 10.1183/13993003.02014-2018
7. Chen K, Pleasants KA, Pleasants RA, et al. Procalcitonin for Antibiotic Prescription in Chronic Obstructive Pulmonary Disease Exacerbations: Systematic Review, Meta-Analysis, and Clinical Perspective. *Pulm Ther.* 2020;6(2):201-214. doi:10.1007/s41030-020-00123-8

Treatment of COPD

- Smoking cessation
- Inhaled bronchodilators, corticosteroids (glucocorticoids), or both
- Supportive care (eg, oxygen therapy, pulmonary rehabilitation)

COPD management involves minimization of symptoms and prevention and treatment of exacerbations (1). Timely diagnosis and initiation of appropriate treatment, particularly smoking cessation, long-term oxygen therapy in patients with hypoxemia, and medications including triple-inhaled therapy (ie, anticholinergics, long-acting beta-agonists, and inhaled corticosteroids (glucocorticoids) in symptomatic patients with frequent exacerbations can reduce long-term mortality. Treatment of cor pulmonale, a common complication of long-standing, severe COPD, is discussed elsewhere.

Smoking cessation is critical in treatment of COPD.

Treatment of chronic stable COPD aims to prevent exacerbations and improve lung and physical function. Treatment relieves symptoms rapidly with primarily short-acting beta-adrenergic medications and decreases exacerbations with inhaled corticosteroids (glucocorticoids), long-acting bronchodilators or a combination (see table Pharmacotherapy of COPD).

Pulmonary rehabilitation includes structured and supervised exercise training, nutrition counseling, and self-management education.

Oxygen therapy is indicated for selected patients.

Treatment of exacerbations ensures adequate oxygenation and near-normal blood pH, reverses airway obstruction, and treats underlying disorders.

Immunization against influenza, pneumococcus, COVID-19, and respiratory syncytial virus (RSV, in patients age 50 years and older) should also be recommended as a preventive measure. Tetanus-diphtheria-pertussis (Tdap) immunization is recommended for patients who were not immunized in adolescence to prevent pertussis.

End-of-life care

In patients with very severe disease, exercise is inadvisable and activities of daily living are arranged to minimize energy expenditure. For example, patients may arrange to live on one floor of the house, have several small meals rather than fewer large meals, and avoid wearing shoes that must be tied. End-of-life care should be discussed, including whether to pursue mechanical ventilation, the use of palliative sedation, and appointment of a surrogate medical decision-maker in the event of the patient’s incapacitation.

Treatment reference

1. Global Initiative for Chronic Obstructive Lung Disease (GOLD): Diagnosis and assessment. Global Strategy for the Prevention, Diagnosis, and Management of COPD: 2026 report.

Prognosis for COPD

Severity of airway obstruction predicts survival in patients with COPD.

Measurement of body mass index (*B*), the degree of airflow obstruction (*O*, which is the FEV_1), dyspnea (*D*, which is measured using the Modified British Medical Research Council (mMRC) Questionnaire), and exercise capacity (*E*, which is measured with a 6-minute walk test) is called the BODE index (1). The BODE index predicts mortality more accurately than FEV_1 alone. Higher scores on the BODE index predict a greater risk of death.

Older age and the presence of heart disease, anemia, resting tachycardia, hypercapnia, or hypoxemia also predict decreased survival, whereas a significant response to bronchodilators predicts improved survival. Risk factors for death in patients with acute exacerbation requiring hospitalization include older age, higher $PaCO_2$, and use of maintenance oral glucocorticoids.

CLINICAL CALCULATORS

BODE Index for COPD Survival Prediction

Patients at high risk of imminent death are those with progressive unexplained weight loss or severe functional decline (eg, those who experience dyspnea with even minimal exertion, such as dressing, bathing, or eating).

Mortality in COPD may result from comorbidities (eg, cardiovascular disease, cancer) or intercurrent illnesses rather than from progression of the underlying disorder in patients who have stopped smoking. Death in patients with COPD is generally due to acute respiratory failure, pneumonia, lung cancer, heart disease, or pulmonary embolism.

Prognosis reference

1. Celli BR, Cote CG, Marin JM, et al. The body-mass index, airflow obstruction, dyspnea, and exercise capacity index in chronic obstructive pulmonary disease. *N Engl J Med.* 2004;350(10):1005-1012. doi:10.1056/NEJMoa021322

Key Points

- **Cigarette smoking in susceptible people is the major cause of chronic obstructive pulmonary disease (COPD) in the resource-rich settings.**
- **COPD should be diagnosed after differentiating from disorders that have similar characteristics (eg, asthma, heart failure) primarily by routine clinical information, such**

as symptoms (particularly time course), age at onset, risk factors, and results of routine tests (eg, chest radiograph, pulmonary function tests).

- Reductions of FEV_1, FVC, and the ratio of FEV_1/FVC post-bronchodilator (also called fixed obstruction) are characteristic findings.
- Patients should be categorized based on symptoms and exacerbation risk into one of 3 groups and use that category to guide treatment.
- Treatment of acute symptoms is primarily with short-acting beta-adrenergic medications.
- Treatment of exacerbations is with inhaled corticosteroids (glucocorticoids), long-acting beta-adrenergic medications, long-acting anticholinergic medications, or a combination.
- The BODE (body mass index, obstruction [FEV_1], dyspnea, exercise capacity) index is helpful in predicting survival in patients with COPD,
- Smoking cessation using multiple interventions should be encouraged in all patients who smoke.
- Immunization against influenza, pneumococcus, COVID-19, Tdap, and RSV (in patients age ≥ 50 years) should also be done as a preventive measure.

Drug Information for the Topic

MERCK MANUAL
Professional Version

Treatment of Acute COPD Exacerbation

Full Review: Jul 2026 By ***Robert A. Wise****, MD, Johns Hopkins Asthma and Allergy Center | Peer reviewed by* ***M. Patricia Rivera****, MD, University of Rochester Medical Center*
Last updated: Jul 2026

Chronic obstructive pulmonary disease (COPD) management involves treatment of chronic stable disease and treatment of exacerbations (1).

Treatment of acute exacerbations involves:

- Oxygen supplementation
- Bronchodilators
- Corticosteroids (glucocorticoids, typically systemic)
- Antibiotics
- Sometimes ventilatory support with noninvasive ventilation or intubation and mechanical ventilation

The immediate objectives of treatment are to ensure adequate oxygenation and near-normal blood pH, reverse airway obstruction, and treat any cause.

The cause of an acute exacerbation is often unknown, although most acute exacerbations result from bacterial or viral infections. Smoking, irritative inhalational exposure, and high levels of air pollution also contribute.

Patients with mild exacerbations and adequate home support can often be treated as outpatients. Older (eg, age over 65 years), frail patients and patients with comorbidities, a history of respiratory failure, or acute changes in blood gas measurements are admitted to the hospital for observation and treatment. Patients with life-threatening exacerbations manifested by uncorrected moderate to severe acute hypoxemia, acute respiratory acidosis, new arrhythmias, or deteriorating respiratory function despite hospital treatment should be admitted to an intensive care unit and their respiratory status monitored frequently.

General reference

1. Global Initiative for Chronic Obstructive Lung Disease (GOLD): Diagnosis and assessment. Global Strategy for the Prevention, Diagnosis, and Management of COPD: 2026 report.

Oxygen in Acute COPD Exacerbation

Many patients require oxygen supplementation during a COPD exacerbation, even those who do not require chronic oxygen support. Hypercapnia may worsen in patients given oxygen. This worsening has traditionally been thought to result from an attenuation of hypoxic respiratory drive. Oxygen administration, even though it may worsen hypercapnia, is recommended. Many patients with COPD have chronic as well as acute hypercapnia and thus severe central nervous system depression is unlikely unless partial pressure of arterial carbon dioxide ($PaCO_2$) is > 85 mm Hg. The target level for partial pressure of arterial oxygen (PaO_2) is approximately 60 mm Hg; higher levels offer little advantage and increase the risk of hypercapnia. In most cases, low concentrations of oxygen will improve hypoxemia. The need for high concentrations of oxygen suggests right to left shunt physiology and other contributors to hypercapnia (eg, pneumonia or pulmonary edema).

In patients who are prone to hypercarbia (ie, an elevated serum bicarbonate may indicate the presence of a compensated respiratory acidosis), oxygen should be given via nasal prongs or Venturi mask so it can be closely regulated, and the patient is closely monitored. Patients whose condition deteriorates during oxygen therapy (eg, those with severe acidemia or central nervous system depression) require additional respiratory support.

Many patients who require oxygen at home for the first time when they are discharged from the hospital after an exacerbation improve within 30 days and no longer require oxygen. Thus, the need for home oxygen should be reassessed 60 to 90 days after discharge.

Ventilatory Support in Acute COPD Exacerbation

Noninvasive positive-pressure ventilation (eg, pressure support or positive airway pressure ventilation by face mask) is an alternative to full mechanical ventilation. Noninvasive ventilation appears to decrease daytime hypercapnia, the need for intubation, hospital stay, and mortality in patients with severe exacerbations (defined as a pH < 7.30 in hemodynamically stable patients not at immediate risk of respiratory arrest) or post-exacerbation (1, 2).

Noninvasive ventilation appears to have no effect in patients with less severe exacerbation. However, it may be indicated for patients with less severe exacerbations whose arterial blood gases (ABGs) worsen despite initial medication or oxygen therapy or who appear to be imminent candidates for full mechanical ventilation but who do not require intubation for control of the airway or sedation for agitation. Patients who have severe dyspnea and hyperinflation and use of accessory muscles of respiration may also gain relief from positive airway pressure. Deterioration while receiving noninvasive ventilation necessitates invasive mechanical ventilation.

Deteriorating ABG values, deteriorating mental status, and progressive respiratory fatigue are indications for endotracheal intubation and mechanical ventilation. Ventilator settings, management strategies, and complications are discussed elsewhere. Risk factors for ventilatory dependence include:

- Forced expiratory volume in 1 second (FEV_1) < 0.5 L
- Stable ABGs with a PaO_2< 50 mm Hg, or a $PaCO_2$> 60 mm Hg
- Severe exercise limitation

- Poor nutritional status

If patients are at high risk, discussion of their wishes regarding intubation and mechanical ventilation should be initiated and documented (see Advance Directives) while their condition is stable and they are not hospitalized. However, overconcern about possible ventilator dependence should not delay management of acute respiratory failure; many patients who require mechanical ventilation can return to their pre-exacerbation level of health.

High-flow nasal oxygen therapy (eg, oxygen administered via nasal cannulation) has also been used for patients with acute respiratory failure due to a COPD exacerbation and can be tried for those who do not tolerate noninvasive mask ventilation.

In patients who require prolonged intubation (eg, > 2 weeks), a tracheostomy is indicated to facilitate comfort, communication, and eating. With a good multidisciplinary pulmonary rehabilitation program, including nutritional and psychologic support, many patients who require prolonged mechanical ventilation can be successfully removed from a ventilator and return to their former level of function.

Specialized programs are available for patients who remain ventilator-dependent after acute respiratory failure. Some patients can remain off the ventilator during the day. For patients with adequate home support, training of family members can permit some patients to be sent home with ventilators.

In patients with severe chronic hypercapnia who were hospitalized for an exacerbation, nocturnal non-invasive ventilation reduces hypercapnia and may improve survival (2). This treatment does not improve long-term quality of life, so implementation should involve shared decision-making with the patient.

Pearls & Pitfalls

- Overconcern about possible ventilator dependence should not delay management of acute respiratory failure; many patients who require mechanical ventilation can return to their pre-exacerbation level of health.

Ventilatory support references

1. Osadnik CR, Tee VS, Carson-Chahhoud KV, Picot J, Wedzicha JA, Smith BJ. Non-invasive ventilation for the management of acute hypercapnic respiratory failure due to exacerbation of chronic obstructive pulmonary disease. *Cochrane Database Syst Rev.* 2017;7(7):CD004104. doi:10.1002/14651858.CD004104.pub4
2. Raveling T, Vonk J, Struik FM, et al. Chronic non-invasive ventilation for chronic obstructive pulmonary disease. *Cochrane Database Syst Rev.* 2021;8(8):CD002878. doi:10.1002/14651858.CD002878.pub3

Medication Therapy in Acute COPD Exacerbation

Beta-agonists and anticholinergics, with or without glucocorticoids, should be started concurrently with oxygen therapy (regardless of how oxygen is administered) with the aim of reversing airway obstruction

(1). Methylxanthines, once considered essential to treatment of acute COPD exacerbations, are no longer used; toxicities exceed benefits.

Beta-agonists

Short-acting beta-agonists (typically albuterol) are the cornerstone of medication therapy for acute exacerbations. Inhalation using a metered-dose inhaler causes rapid bronchodilation; there are no data indicating that doses taken with nebulizers are more effective than the same doses correctly taken with metered-dose inhalers. In cases of severe, unresponsive bronchospasm, continuous nebulizer treatments may sometimes be administered.

Anticholinergic agents

Ipratropium, a short-acting anticholinergic, is effective in acute COPD exacerbations and should be given concurrently or alternating with beta-agonists. Dosage is 0.25 to 0.5 mg by nebulizer or 2 inhalations (17 to 18 mcg of drug delivered per puff) by metered-dose inhaler every 4 to 6 hours. Ipratropium generally provides a bronchodilating effect similar to that of usual recommended doses of beta-agonists.

The use of longer-acting anticholinergics for the treatment of acute COPD exacerbations has not been studied, and their use is generally reserved for the exacerbation prevention (or maintenance therapy).

Glucocorticoids

Systemic glucocorticoids should be initiated immediately for all but mild exacerbations. Options include prednisone 30 to 60 mg orally once a day for 5 to 7 days and stopped directly or tapered over 7 to 14 days depending on the clinical response. A parenteral alternative is methylprednisolone 60 to 500 mg IV once a day for 3 days and then tapered over 7 to 14 days. These medications are equivalent in their acute effects.

Antibiotics

Routine cultures and Gram stains are not necessary before treatment unless an unusual or resistant organism is suspected (eg, in patients who are hospitalized, institutionalized, or immunosuppressed). Medications directed against oral flora are indicated.

Antibiotics should be initiated if any of the following conditions are met:

- Presence of three cardinal symptoms: Increase in dyspnea, increase in sputum volume, and increase in sputum purulence.
- Presence of two cardinal symptoms: Increased sputum purulence (yellow or green sputum), along with increases in dyspnea or sputum volume
- Patients requiring invasive or noninvasive ventilation

The choice of antibiotic is dictated by local patterns of bacterial sensitivity and patient history. Macrolides (eg, azithromycin), amoxicillin (usually with clavulanic acid), or tetracyclines (eg, doxycycline) is given for 7 to 14 days.

When patients are seriously ill or clinical evidence suggests that the infectious organisms are resistant, broader spectrum second-line medications can be used. These medications include fluoroquinolones (eg, ciprofloxacin, levofloxacin), and second-generation cephalosporins (eg, cefuroxime, cefaclor). These medications are effective against beta-lactamase–producing strains of *Haemophilus influenzae* and *Moraxella catarrhalis* but have not been shown to be more effective than first-line drugs for most patients.

Patients can be taught to recognize a change in sputum from normal to purulent as a sign of impending exacerbation and to start a 10- to 14-day course of antibiotic therapy.

Testing for low C-reactive protein (CRP) levels may guide which patients can avoid use of antibiotics (2).

Long-term antibiotic prophylaxis is recommended primarily for patients with underlying structural changes in the lung, such as bronchiectasis or infected bullae.

Other medications

Antitussives, such as dextromethorphan and benzonatate, have no role in treating COPD exacerbations.

Opioids (eg, codeine, hydrocodone, oxycodone) should be used judiciously for relief of symptoms (eg, severe coughing paroxysms, pain) because these medications may suppress a productive cough, impair mental status, and cause constipation.

Medication therapy references

1. Global Initiative for Chronic Obstructive Lung Disease (GOLD): Diagnosis and assessment. Global Strategy for the Prevention, Diagnosis, and Management of COPD: 2026 report.
2. Butler CC, Gillespie D, White P, et al. C-Reactive Protein Testing to Guide Antibiotic Prescribing for COPD Exacerbations. *N Engl J Med.* 2019;381(2):111-120. doi:10.1056/NEJMoa1803185

Key Points

- **Most patients hospitalized with exacerbation of chronic obstructive pulmonary disease (COPD) require oxygen supplementation during an exacerbation.**
- **Inhaled short-acting beta-agonists are the cornerstone of medication therapy for acute exacerbations.**
- **Use antibiotics if patients have acute exacerbations and purulent sputum with other suggestive symptoms, are mechanically ventilated, or have had prior positive sputum culture from another past exacerbation.**
- **For patients with severe COPD, address end-of-life care proactively, including preferences regarding mechanical ventilation and palliative sedation.**

Drug Information for the Topic

MERCK MANUAL
Professional Version

Treatment of Stable COPD

*Full Review: Jul 2026 By **Robert A. Wise**, MD, Johns Hopkins Asthma and Allergy Center | Peer reviewed by **M. Patricia Rivera**, MD, University of Rochester Medical Center*
Last updated: Jul 2026

Chronic obstructive pulmonary disease (COPD) management involves treatment of chronic stable COPD and treatment of exacerbations (1).

Treatment of chronic stable COPD aims to prevent exacerbations and improve lung and physical function through:

- Smoking cessation
- Medications
- Oxygen therapy
- Enhancement of nutrition
- Pulmonary rehabilitation, including exercise

Surgical treatment of COPD is indicated for selected patients. This may include lung volume reduction procedures or lung transplantation.

General reference

1. Global Initiative for Chronic Obstructive Lung Disease (GOLD): Diagnosis and assessment. Global Strategy for the Prevention, Diagnosis, and Management of COPD: 2026 report.

Smoking Cessation in Stable COPD

Smoking cessation is difficult but extremely important; it slows but does not halt the rate of decline of the volume of air forcefully expired during the first second after taking a full breath (FEV_1) (see figure Changes in Lung Function in Patients Who Quit Smoking) and increases long-term survival. Simultaneous use of multiple strategies is most effective:

- Establishment of a quit date
- Behavior modification techniques
- Group sessions
- Nicotine replacement therapy (by gum, transdermal patch, inhaler, lozenge, or nasal spray)
- Varenicline or bupropion

- Physician encouragement

Quit rates are generally low among patients with COPD (1), even with the most effective interventions, such as the use of bupropion combined with nicotine replacement or use of varenicline alone.

Changes in Lung Function (Percentage of Predicted FEV_1) in Patients Who Quit Smoking Compared With Those Who Continue

During the first year, lung function improved in patients who quit smoking and declined in those who continued. Subsequently, the rate of decline in those who continued was twice that of those who quit. Function declined in those who relapsed and improved in those who quit regardless of when the change occurred. Based on data from Scanlon PD, Connett JE, Waller LA, et al. Smoking cessation and lung function in mild-to-moderate chronic obstructive pulmonary disease; the Lung Health Study. *Am J Respir Crit Care Med.* 2000;161(2 Pt 1):381-390. doi:10.1164/ajrccm.161.2.9901044

Smoking cessation reference

1. Liu Y, Greenlund KJ, VanFrank B, et al. Smoking Cessation Among U.S. Adult Smokers With and Without Chronic Obstructive Pulmonary Disease, 2018. *Am J Prev Med.* 2022;62(4):492-502. doi:10.1016/j.amepre.2021.12.001

Pharmacotherapy for Stable COPD

Recommended medications for COPD are summarized in the table Pharmacotherapy of COPD.

Inhaled bronchodilators are the mainstay of COPD management; medications include:

- Beta-agonists
- Anticholinergics (antimuscarinics)

These 2 medication classes are equally effective. Patients with mild disease are treated only when symptomatic. Patients with moderate to severe COPD should be taking medications from one or both of these classes regularly to improve pulmonary function and increase exercise capacity. The initial choice among short-acting beta-agonists, long-acting beta-agonists, anticholinergics, and combination beta-agonist and anticholinergic therapy is influenced by factors such as cost, convenience, and patient preference as well as therapeutic response.

Breathing with COPD

The frequency of exacerbations can be reduced with the use of anticholinergics, long-acting beta-agonists, and inhaled corticosteroids (glucocorticoids). However, there is no convincing evidence that regular bronchodilator use slows deterioration of lung function.

For home treatment of chronic stable disease, medication administration by metered-dose inhaler or dry-powder inhaler is preferred over administration by nebulizer; home nebulizers are prone to contamination when cleaning and drying are incomplete. Nebulizers deliver medication to more proximal airways due to tidal breathing compared to total lung capacity maneuvers with inhalers, leading to inefficient medication administration. Therefore, nebulizers should be reserved for people who cannot coordinate activation of the metered-dose inhaler with inhalation or cannot develop enough inspiratory flow for dry powder inhalers.

How to Use a Metered-Dose Inhaler With a Spacer

- Shake the inhaler after removing the caps from the inhaler and the spacer.
- Attach the spacer to the inhaler.
- Exhale fully for 1 or 2 seconds. Try to get as much air out of your lungs as you can.
- Put the spacer between your teeth and close your lips tightly around it.
- Breathe in slowly through your mouth.
- Press the top of the inhaler and keep breathing slowly and deeply.
- Take the spacer out of your mouth.
- Hold your breath for 10 seconds (or as long as you can).
- Breathe out and, if a second dose is required, repeat the process after 1 minute.
- Put the caps back on the inhaler and the spacer.

For metered-dose inhalers, patients should be taught to exhale to functional residual capacity, inhale the aerosol for approximately 3 to 4 seconds to total lung capacity, and hold the inhalation for 10 seconds (or as long as is possible) before exhaling. Spacers help ensure optimal delivery of medication to the distal airways and reduce the importance of coordinating activation of the inhaler with inhalation. Some spacers alert patients if they are inhaling too rapidly. New or not recently used metered-dose inhalers require 2 to 3 priming doses (different manufacturers have slightly different recommendations for what is considered "not recently used," ranging from 3 to 14 days).

Dry powder inhalers, in contrast to metered-dose inhalers, require a rapid deep inspiratory effort in order to disaggregate the powder so that the smallest particles are deposited in the terminal airspaces.

TABLE

Pharmacotherapy for COPD

Approach to Therapy	Dyspnea Without Exacerbations	Exacerbations With or Without Dyspnea
Initial therapy*	LABA or LAMA	LABA or LAMA
Initial escalation of therapy	LABA + LAMA	LABA + LAMA or LABA + LAMA + ICS†
Subsequent approach to therapy for persistent symptoms	Consider changing inhaler device or medication Implement or escalate non-pharmacological treatments Evaluation for other causes of dyspnea	LABA + LAMA + ICS + roflumilast or azithromycin‡

* Rescue short-acting bronchodilators should be prescribed to all patients for immediate symptom relief.

† ICS is most effective in patients with eosinophilia ≥ 150/mcL (≥ 0.15 × 10^9/L). May need to de-escalate ICS if the patient develops pneumonia or does not respond to treatment. In patients with eosinophilia ≥ 300 mcL (≥ 0.30 × 10^9/L), de-escalation is more likely to cause an exacerbation.

‡ For recurrent exacerbations, add either a phosphodiesterase-4 (PDE4) inhibitor (eg, roflumilast) or long-term azithromycin. Roflumilast is indicated in patients with COPD (FEV_1< 50%) with chronic bronchitis. Azithromycin is preferred for former smokers and is less effective in current smokers. If exacerbations persist, patients with eosinophilia may benefit from a biologic such as mepolizumab or dupilumab.

FEV_1 = The volume of air forcefully expired during the first second after taking a full breath; LABA = long-acting beta agonist; LAMA = long-acting antimuscarinic antagonist; ICS = inhaled corticosteroid (glucocorticoid).

Adapted from the Global Initiative for Chronic Obstructive Lung Disease (GOLD): Global Strategy for the Prevention, Diagnosis, and Management of COPD: 2026 report. Available at http://www.goldcopd.org.

Beta-agonists

Beta-agonists, which can be short-acting or long-acting, relax bronchial smooth muscle and increase mucociliary clearance.

Albuterol is a **short-acting beta-agonist** aerosol, dosed as 2 puffs inhaled from a metered-dose inhaler 4 to 6 times a day as needed. It is usually the medication of choice for episodic dyspnea or early treatment of exacerbations.

Long-acting beta-agonists are preferable for patients with nocturnal symptoms or for those who find frequent dosing inconvenient. They may be used as initial therapy for COPD in some patients; however, a long-acting beta-agonist is typically used together with a long-acting muscarinic antagonist. Options are typically used at the same time each day and include salmeterol powder 1 puff (50 mcg) inhaled twice a day, indacaterol 1 puff (75 mcg) inhaled once a day (150 mcg or 300 mcg once a day in Europe), and olodaterol 2 puffs (5 mcg) once a day. Vilanterol and formoterol are available in combination with an inhaled corticosteroid (glucocorticoid). Also available are nebulized forms of arformoterol and formoterol. The dry-powder formulations may be more effective for patients who have trouble coordinating use of a metered-dose inhaler.

Patients should be taught the difference between short-acting and long-acting medications, because long-acting medications that are used as needed or more than twice a day increase the risk of cardiac arrhythmias.

Adverse effects commonly include tremor, anxiety, tachycardia, and mild, temporary hypokalemia; adverse effects are similar for both long-acting and short-acting beta-agonists.

Anticholinergics

Long-acting muscarinic antagonists are anticholinergics (sometimes called antimuscarinics)that relax bronchial smooth muscle through competitive inhibition of muscarinic receptors (M1 and M3). Long-acting muscarinic antagonists are a reasonable alternative as first-line therapy for patients with COPD. Adverse effects of all anticholinergics are pupillary dilation (and risk of triggering or worsening acute angle-closure glaucoma), urinary retention, and dry mouth.

Tiotropium is a long-acting quaternary anticholinergic inhaled as a powder formulation (18 mcg/puff) or as a soft mist inhaler (2.5 mcg/puff). Dose is 1 puff (18 mcg) once a day by dry powder inhaler and 2 puffs (5 mcg) once a day by soft mist inhaler.

Aclidinium is available as a multidose dry-powder inhaler. Dose is 1 puff (400 mcg/puff) twice a day. Aclidinium is also available in combination with a long-acting beta-agonist as a dry-powder inhaler.

Umeclidinium can be used as a once a day combination with vilanterol (a long-acting beta-agonist) in a dry-powder inhaler.

Glycopyrrolate (an anticholinergic) can be used twice a day in combination with indacaterol or formoterol (long-acting beta-agonists) in a dry-powder or metered-dose inhaler.

Revefenacin is given once a day via nebulizer.

Ipratropium is a **short-acting anticholinergic**; dose is 2 puffs (17 mcg/puff) from a metered-dose inhaler every 4 to 6 hours not to exceed 12 puffs a day. Ipratropium has a slower onset of action compared to a beta-agonist (within 30 minutes; peak effect in 1 to 2 hour), so a beta-agonist is often prescribed with it in a single combination inhaler or as a separate as-needed rescue medication.

Inhaled corticosteroids (glucocorticoids)

Glucocorticoids are often part of the treatment of COPD (1, 2, 3). Inhaled corticosteroids (glucocorticoids) may reduce airway inflammation, reverse beta-receptor down-regulation, and inhibit leukotriene and cytokine production. However, they have not been shown to alter the course of pulmonary function decline in patients with COPD who continue to smoke. Glucocorticoids relieve symptoms and improve short-term pulmonary function in some patients, are additive to the effect of bronchodilators, and diminish the frequency of COPD exacerbations. They are indicated for patients who have repeated exacerbations or symptoms despite optimal bronchodilator therapy. Inhaled glucocorticoids are of greatest benefit in patients who have eosinophilia (1). In patients with exacerbations, the addition of an inhaled corticosteroid reduces exacerbation frequency and prolongs survival (2, 3).

The long-term risks of inhaled glucocorticoids in older adults are not proved but probably include osteoporosis, cataract formation, and an increased risk of nonfatal pneumonia. Long-term users therefore should undergo periodic ophthalmologic and bone densitometry screening if indicated (eg, by a history of changes in night vision, frailty, or bone fractures) and should possibly receive supplemental calcium, vitamin D, and a bisphosphonate as indicated.

De-escalation of inhaled corticosteroid (glucocorticoid) therapy may be necessary if patients develop pneumonia or do not respond to treatment.

Use of a combination of a long-acting beta-agonist and an inhaled corticosteroid without the use of a long-acting anticholinergic is not recommended for patients with COPD unless they have concomitant asthma.

Oral or other systemic (eg, IV) glucocorticoids are typically not used to treat chronic stable COPD.

Phosphodiesterase inhibitors

Phosphodiesterase-4 (PDE4) inhibitors such as roflumilast are more specific than the non-selective phosphodiesterase inhibitor theophylline and have fewer adverse effects. They have anti-inflammatory properties and are mild bronchodilators. Phosphodiesterase-4 inhibitors can be used in addition to other bronchodilators for reduction of exacerbations in patients who have symptomatic COPD with chronic bronchitis.

Common adverse effects include nausea, headache, and weight loss, but these effects are minimized if the medication is started at a low dose and increased over several weeks.

Ensifentrine is an inhaled selective dual phosphodiesterase (PDE3) and PDE4 inhibitor that has both bronchodilator and anti-inflammatory effects can be used as an add-on agent for patients who are using other bronchodilators. It may reduce exacerbation risk and is generally well tolerated (4).

Macrolide antibiotics

Long-term azithromycin therapy is an effective adjunct to prevent COPD exacerbations in patients who are prone to repeated or severe exacerbations, particularly those who are not currently smoking. An azithromycin dose of 250 mg orally once a day has demonstrated efficacy (5). Azithromycin may also be administered 500 mg orally 3 times a week (6). Erythromycin is an alternative. Macrolides do not slow disease progression.

Theophylline

Theophylline, a non-selective phosphodiesterase inhibitor, is not recommended for the treatment of chronic stable COPD because safer, more effective medications are available. Theophylline decreases smooth muscle spasm, enhances mucociliary clearance, improves right ventricular function, and decreases pulmonary vascular resistance and arterial pressure. Its mode of action is poorly understood but appears to differ from that of beta-2-agonists and anticholinergics. Its role in improving diaphragmatic function and dyspnea during exercise is unclear. Toxicity is common and includes sleeplessness and gastrointestinal upset. With high blood levels, ventricular arrhythmias and seizures can occur.

Pharmacotherapy references

1. Pascoe S, Barnes N, Brusselle G, et al. Blood eosinophils and treatment response with triple and dual combination therapy in chronic obstructive pulmonary disease: analysis of the IMPACT trial. *Lancet Respir Med.* 2019;7(9):745-756. doi: 10.1016/S2213-2600(19)30190-0
2. Lipson DA, Crim C, Criner GJ, et al. Reduction in all-cause mortality with fluticasone furoate/umeclidinium/vilanterol in patients with chronic obstructive pulmonary disease. *Am J Respir Crit Care Med.* 2020;201(12):1508-1516. doi: 10.1164/rccm.201911-2207OC
3. Rabe KF, Martinez FJ, Ferguson GT, et al. Triple inhaled therapy at two glucocorticoid doses in moderate-to-very-severe COPD. *N Engl J Med.* 2020;383(1):35-48. doi: 10.1056/NEJMoa1916046
4. Anzueto A, Barjaktarevic IZ, Siler TM, et al. Ensifentrine, a Novel Phosphodiesterase 3 and 4 Inhibitor for the Treatment of Chronic Obstructive Pulmonary Disease: Randomized, Double-Blind, Placebo-controlled, Multicenter Phase III Trials (the ENHANCE Trials). *Am J Respir Crit Care Med.* 2023;208(4):406-416. doi:10.1164/rccm.202306-0944OC
5. Albert RK, Connett J, Bailey WC, et al: Azithromycin for prevention of exacerbations of COPD. *N Engl J Med.* 2011;365(8):689-698. doi:10.1056/NEJMoa1104623
6. Global Initiative for Chronic Obstructive Lung Disease (GOLD): Diagnosis and assessment. Global Strategy for the Prevention, Diagnosis, and Management of COPD: 2026 report.

Oxygen Therapy for Stable COPD

Long-term oxygen therapy prolongs life in patients with COPD who have severe resting hypoxemia (ie, partial pressure of arterial oxygen [PaO_2] chronically ≤ 55 mm Hg or oxygen saturation [SpO_2] ≤ 88%) (1). Patients should be instructed to use oxygen for at least 15 hours per day. Oxygen therapy works by:

- Bringing hematocrit toward normal levels
- Improving neuropsychologic factors, possibly by facilitating sleep
- Ameliorating pulmonary hemodynamic abnormalities
- Increasing exercise tolerance in some patients

Oxygen saturation should be measured during exercise and while the patient is at rest. Similarly, a sleep study should be considered for patients with advanced COPD who do not meet the criteria for long-term oxygen therapy while they are awake (see table Indications for Long-Term Oxygen Therapy in COPD) but whose clinical assessment suggests pulmonary hypertension in the absence of daytime hypoxemia. Nocturnal oxygen may be prescribed if a sleep study shows episodic oxygen desaturation to ≤ 88%. Such treatment prevents progression of pulmonary hypertension, but its effects on survival are unknown. In patients with moderate hypoxemia above 88% or exercise desaturation, oxygen therapy may reduce symptoms, but there is no improvement in survival or reduction in hospitalizations (2).

Some patients need supplemental oxygen during air travel because flight cabin pressure in commercial airliners is below sea level air pressure (often equivalent to 1830 to 2400 m [6000 to 8000 ft]). Patients with COPD who are eucapnic and have a PaO_2> 68 mm Hg at sea level generally have an in-flight PaO_2> 50 mm Hg and do not require supplemental oxygen. All patients with COPD with a PaO_2≤ 68 mm Hg at sea level, hypercapnia, significant anemia (hematocrit < 30), or a coexisting heart or cerebrovascular disorder should use supplemental oxygen during long flights and should notify the airline when making their reservation. Because portable oxygen concentrators are ubiquitous, very few airlines still provide oxygen. Patients are advised to rent or purchase a portable oxygen concentrator for use during air travel, including enough spare batteries to account for the flight time and possible delays.

All patients must be taught the dangers of smoking or cooking over an open flame during oxygen use because oxygen accelerates combustion.

TABLE

Indications for Long-Term Oxygen Therapy in COPD

PaO_2≤ 55 mm Hg or SaO_2 ≤ 88%* in patients receiving optimal medical regimen for at least 30 days†
PaO_2= 55 to 59 mm Hg or SaO_2≤ 89%* for patients with cor pulmonale or erythrocytosis (> 55%)
Can be considered for patients with exercise desaturation or severe dyspnea if there is symptomatic improvement; however, there is no improvement in survival or hospitalization. May also be considered for patients with nocturnal desaturation.‡

* Arterial oxygen levels are measured at rest during air breathing.

† Patients who are recovering from an acute respiratory illness and who meet the listed criteria should be given oxygen and rechecked while breathing room air after 60 to 90 days.

‡ See also Long-Term Oxygen Treatment Trial Research Group, Albert RK, Au DH, et al. A randomized trial of long-term oxygen for COPD with moderate desaturation. *N Engl J Med.* 2016;375(17):1617-1627. doi:10.1056/NEJMoa1604344

Hct = hematocrit; PaO_2 = arterial oxygen partial pressure; SaO_2 = oxygen saturation.

Oxygen administration

Oxygen is administered by nasal cannula at a flow rate sufficient to achieve a PaO_2> 60 mm Hg (oxygen saturation > 90%), usually ≤ 3 L/minute at rest (frequently higher at higher altitudes).

Oxygen is supplied by electrically driven oxygen concentrators, or cylinders of compressed gas. Stationary concentrators, which limit mobility but are the least expensive, are preferable for patients who spend most of their time at home. Such patients require small oxygen tanks for backup in case of an electrical failure and for portable use. Portable concentrators that allow mobility can be used for patients who do not require high flow rates.

Large compressed oxygen cylinders are the most expensive way of providing oxygen and should be used only if no other source is available.

Various oxygen-conserving devices can reduce the amount of oxygen used by the patient, either by using a reservoir system or by permitting oxygen flow only during inspiration. Systems with these devices may correct resting hypoxemia as effectively as continuous flow devices. However, intermittent flow devices may not be as effective as continuous flow for exercise-associated hypoxemia.

Oxygen therapy references

1. Cagle SD Jr, Landrum LS, Kennedy AM. Chronic Obstructive Pulmonary Disease: Diagnosis and Management. *Am Fam Physician.* 2023;107(6):604-612.
2. Long-Term Oxygen Treatment Trial Research Group. A randomized trial of long-term oxygen for COPD with moderate desaturation. *N Engl J Med.* 2016;375(17):1617-1627. doi:10.1056/NEJMoa1604344

Vaccinations in Stable COPD

All patients with COPD should be given annual influenza vaccinations. If COPD increases the risk of respiratory complications from influenza, prophylactic treatment with an anti-influenza drug (baloxavir, oseltamivir or zanamivir) is recommended after close exposure to influenza-infected people. Treatment with a neuraminidase inhibitor or an endonuclease inhibitor (baloxavir) should be started at the first sign of an influenza-like illness and within 2 days of the exposure.

Pneumococcal vaccines should be administered because they are effective in reducing community-acquired pneumonia and invasive pneumococcal infections in older adults.

Respiratory syncytial virus (RSV) vaccine is recommended for adults over the age of 60 years regardless of COPD diagnosis, but COPD patients are likely to have significant benefit from RSV vaccination.

COVID-19 vaccination is also recommended for patients with COPD.

Diphtheria-tetanus-pertussis vaccine (Tdap) should be administered to protect against pertussis for those patients who did not receive immunization in adolescence. Adults should receive the Tdap or tetanus-diphtheria (Td) vaccines every 10 years to maintain protection against diphtheria.

Nutrition in Stable COPD

Patients with COPD are at risk of weight loss and nutritional deficiencies because of a higher energy cost of daily activities, reduced caloric intake relative to need because of dyspnea, and the catabolic effect of inflammatory cytokines such as tumor necrosis factor (TNF)-alpha. Generalized muscle strength and efficiency of oxygen use are impaired. Patients with poorer nutritional status have a worse prognosis, so it is prudent to recommend a balanced diet with adequate caloric intake in conjunction with exercise to prevent or reverse undernutrition and muscle atrophy. Split-meal feeding can be helpful and leads to improved body composition and small increases in functional capacity.

Nutritional supplementation should be limited to patients who are undernourished, who may experience benefits in respiratory muscle strength and quality of life with from such intervention (1). Overnutrition (ie, excessive weight gain) should be avoided, and patients who have obesity should strive to gradually reduce body fat.

The use of appetite stimulants (eg, megestrol acetate), anabolic steroids, growth hormone supplementation, and TNF antagonists has failed to yield clinically meaningful improvement in outcomes (ie, reversing undernutrition as well as simultaneously improving functional status and prognosis) in patients with COPD (2, 3, 4, 5)

Nutrition references

1. Huang WJ, Ko CY. Systematic review and meta-analysis of nutrient supplements for treating sarcopenia in people with chronic obstructive pulmonary disease. *Aging Clin Exp Res.* 2024;36(1):69.

doi:10.1007/s40520-024-02722-w
2. Burdet L, de Muralt B, Schutz Y, Pichard C, Fitting JW. Administration of growth hormone to underweight patients with chronic obstructive pulmonary disease. A prospective, randomized, controlled study. *Am J Respir Crit Care Med.* 1997;156(6):1800-1806. doi:10.1164/ajrccm.156.6.9704142
3. Herrejón A, Palop J, Inchaurraga I, et al. Dosis bajas de acetato de megestrol aumentan el peso y mejoran la nutrición de los pacientes con enfermedad pulmonar obstructiva crónica grave y pérdida de peso [Low doses of megestrol acetate increase weight and improve nutrition status in patients with severe chronic obstructive pulmonary disease and weight loss]. *Med Clin (Barc).* 2011;137(5):193-198. doi:10.1016/j.medcli.2011.02.016
4. Liu Y, Huang C, Du J, et al. Anabolic-androgenic steroids for patients with chronic obstructive pulmonary disease: A systematic review and meta-analysis. *Front Med (Lausanne).* 2022;9:915159. doi:10.3389/fmed.2022.915159
5. Matera MG, Calzetta L, Cazzola M. TNF-alpha inhibitors in asthma and COPD: we must not throw the baby out with the bath water. *Pulm Pharmacol Ther.* 2010;23(2):121-128. doi:10.1016/j.pupt.2009.10.007

Pulmonary Rehabilitation in Stable COPD

Pulmonary rehabilitation programs serve as adjuncts to medication treatment to improve physical function; many hospitals and health care organizations offer formal multidisciplinary rehabilitation programs. Pulmonary rehabilitation includes:

- Exercise
- Education
- Behavioral interventions

Treatment should be individualized; patients and family members are taught about COPD and medical treatments, and patients are encouraged to take as much responsibility for personal care as possible.

The benefits of rehabilitation are greater independence and improved quality of life and exercise capacity. Pulmonary rehabilitation typically does not improve pulmonary function. A carefully integrated rehabilitation program helps patients with severe COPD accommodate to physiologic limitations while providing realistic expectations for improvement. Patients with severe disease require a minimum of 3 months of rehabilitation to benefit and should continue with maintenance programs.

An exercise program can be successfully performed in many settings (eg, the home, in the hospital, in other institutional settings).

Graded exercise can ameliorate skeletal muscle deconditioning resulting from inactivity or prolonged hospitalization for respiratory failure. Specific training of respiratory muscles is less helpful than general aerobic conditioning.

A typical training program begins with slow walking on a treadmill or unloaded cycling on an ergometer for a few minutes. Duration and exercise load are progressively increased over 4 to 6 weeks until the patient can exercise for 20 to 30 minutes nonstop with manageable dyspnea. Patients with very severe COPD can usually achieve an exercise regimen of walking for 30 minutes at 1 to 2 miles per hour (1.6 to 3.2 km per hour). Maintenance exercise should be done 3 to 4 times/week to maintain fitness levels. Oxygen saturation is monitored, and supplemental oxygen is provided as needed.

Upper extremity resistance training helps the patient in doing daily tasks (eg, bathing, dressing, house cleaning). The usual benefits of exercise are modest increases in lower extremity strength, endurance, and maximum oxygen consumption.

Patients should be taught ways to conserve energy during activities of daily living and to pace their activities. Difficulties in sexual function should be discussed and advice should be given on using energy-conserving techniques for sexual gratification.

Pulmonary rehabilitation may also be successfully implemented using telehealth models with variable types of access and supervision (1, 2). In general, these programs lead to similar improvements in exercise capacity and have been associated with improvements in lung function (FEV_1, FEV_1/forced vital capacity) and compliance. Because of the potential for difficulty in accessing formal in-person pulmonary rehabilitation programs, it is recommended that virtual rehabilitation be offered to patients as an alternative to an in-person program.

Pulmonary rehabilitation references

1. Qu M, Yang L, Liu K, et al. The effectiveness and feasibility of virtual pulmonary rehabilitation in patients with chronic respiratory diseases: A systematic review and meta-analysis. *Respir Med.* 2025;244:108159. doi:10.1016/j.rmed.2025.108159
2. Rochester CL, Alison JA, Carlin B, et al. Pulmonary Rehabilitation for Adults with Chronic Respiratory Disease: An Official American Thoracic Society Clinical Practice Guideline. *Am J Respir Crit Care Med.* 2023;208(4):e7-e26. doi:10.1164/rccm.202306-1066ST

Surgery in Stable COPD

Surgical options for treatment of severe COPD include:

- Lung volume reduction
- Lung transplantation

Lung volume reduction surgery

Lung volume reduction surgery consists of resecting nonfunctioning emphysematous areas. The procedure improves lung function, exercise tolerance, and quality of life in patients with severe, predominantly upper-lung emphysema who have low baseline exercise capacity after pulmonary rehabilitation (1).

The effect on arterial blood gas measurements is variable and not predictable, but most patients who require oxygen therapy before surgery continue to need it. Improvement is less than that with lung transplantation. The mechanism of improvement is believed to be enhanced lung recoil and improved diaphragmatic function.

The best candidates for lung volume reduction surgery are patients with:

- FEV_1 of 20 to 40% of predicted
- Diffusing capacity for carbon monoxide (DLCO) > 20% of predicted
- $PaCO_2$< 50 mm Hg
- Significantly impaired exercise capacity
- Heterogeneous pulmonary disease on CT with an upper-lobe predominance

Rarely, patients have extremely large bullae that compress the functional lung. These patients can be helped by surgical resection of these bullae, with resulting relief of symptoms and improved pulmonary function. Generally, resection is most beneficial for patients with bullae affecting more than one-third of a hemithorax and an FEV_1 approximately half of the predicted normal value. Improved pulmonary function is related to the amount of normal or minimally diseased lung tissue that was compressed by the resected bullae. Serial chest radiographs and CT scans are the most useful procedures for determining whether a patient's functional status is due to compression of viable lung by bullae or to generalized emphysema. A markedly reduced DLCO (< 40% predicted) indicates widespread emphysema and suggests a poorer outcome from surgical resection.

Patients with severe pulmonary hypertension (systolic pulmonary artery pressure [sPAP] ≤ 45 mm Hg) or significant coronary artery disease are not ideal candidates for lung volume reduction surgery.

Endobronchial valve placement

Placement of an endobronchial valve is an alternative to lung volume reduction surgery in some patients. Endobronchial valves that are inserted with a bronchoscope and cause atelectasis of emphysematous target regions of the lung are available in selected centers for selected patients who do not have collateral ventilation to these regions (2). Surgical lung volume reduction is usually bilateral, whereas bronchoscopic procedures are usually unilateral and therefore result in less improvement.

Lung transplantation

Lung transplantation can be single or double. Perioperative complications tend to be lower with single-lung transplantation, but some evidence shows that survival time is increased with double-lung transplantation (3). Candidates for transplantation are patients < 65 years with an FEV_1< 25% predicted after bronchodilator therapy or with severe pulmonary hypertension. The goal of lung transplantation is to improve quality of life, because survival time is not necessarily increased. Lifelong immunosuppression is required, with the attendant risk of opportunistic infections.

Surgery references

1. Fishman A, Martinez F, Naunheim K, et al. A randomized trial comparing lung-volume-reduction surgery with medical therapy for severe emphysema. *N Engl J Med.* 2003;348(21):2059-2073. doi:10.1056/NEJMoa030287

2. Criner GJ, Sue R, Wright S, et al. A Multicenter Randomized Controlled Trial of Zephyr Endobronchial Valve Treatment in Heterogeneous Emphysema (LIBERATE). *Am J Respir Crit Care Med.* 2018;198(9):1151-1164. doi:10.1164/rccm.201803-0590OC

3. Yu H, Bian T, Yu Z, et al. Bilateral Lung Transplantation Provides Better Long-term Survival and Pulmonary Function Than Single Lung Transplantation: A Systematic Review and Meta-analysis. *Transplantation.* 2019;103(12):2634-2644. doi:10.1097/TP.0000000000002841

Key Points

- **Relieve symptoms of COPD rapidly with primarily short-acting beta-adrenergic medications and decrease exacerbations with inhaled corticosteroids (glucocorticoids), long-acting beta-adrenergic medications, long-acting antimuscarinic medications, or a combination.**
- **Encourage smoking cessation using multiple interventions (eg, behavior modification, support groups, nicotine replacement, varenicline therapy).**
- **Optimize use of supportive treatments (eg, nutrition, pulmonary rehabilitation, self-directed exercise).**

Drug Information for the Topic